generators, $\mathcal{Y}_{ij}$, have all zero entries except for the elements defined by ij, such that $\mathcal{Y}_{ij} = -\mathcal{Y}_{ji} = -i$. Examples are:

$$\mathcal{X}_{13} = \begin{pmatrix} 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{pmatrix} \ , \ \mathcal{Y}_{13} \ = \ \begin{pmatrix} 0&0&-i&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ i&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{pmatrix} \ . \tag{C3}$$

Appendix D: Contractions of Color-Charge Operators

Expressions for the contractions of color charge-charge operators can be found straightforwardly. Acting on a qu8it lattice site twice (for a 2-site system), the summations over adjoint indices reduce to

$$\begin{aligned} \sum_a \left(\widetilde{Q}^{(a)} \otimes \widetilde{I}\right)^2 &= \frac{4}{3}\mathrm{diag}\,(0,1,1,1,1,1,1,0) \otimes \widetilde{I} \\ &= 8w_1 \otimes w_1 - \frac{8}{3}\left(w_3 + w_5 + w_7\right) \otimes w_1 \\ &= \widetilde{I} \otimes \widetilde{I} - \mathrm{diag}\left(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \otimes \widetilde{I} \ , \end{aligned} \tag{D1}$$

and acting on a anti-qu8it lattice site twice, the summations over adjoint indices reduce to

$$\begin{aligned} \sum_a \left(\widetilde{I} \otimes \widetilde{\bar{Q}}^{(a)}\right)^2 &= \frac{4}{3}\widetilde{I} \otimes \mathrm{diag}\,(0,1,1,1,1,1,1,0) \\ &= 8w_1 \otimes w_1 - \frac{8}{3} w_1 \otimes \left(w_3 + w_5 + w_7\right) \\ &= \widetilde{I} \otimes \widetilde{I} - \widetilde{I} \otimes \mathrm{diag}\left(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \ . \end{aligned} \tag{D2}$$

In general, there will be contributions from color-charge operators acting on two sites, of the form quark-quark, anti-quark-anti-quark and quark-anti-quark. For color-charge operators acting on arbitrary quark-quark and anti-quark-anti-quark sites,

$$\begin{aligned} \sum_a \widetilde{Q}^{(a)} \otimes \widetilde{Q}^{(a)} \ = \ \sum_a \widetilde{\bar{Q}}^{(a)} \otimes \widetilde{\bar{Q}}^{(a)} \ &= \ \frac{1}{2}\left(w_3 - w_5\right) \otimes \left(w_3 - w_5\right) \ + \ \frac{1}{6}\left(w_4 + w_6 - 2w_8\right) \otimes \left(w_4 + w_6 - 2w_8\right) \\ &+ \frac{1}{4} \sum_{\substack{r \in \{(23),(24),(34),\\ (56),(57),(67)\}}} \left(\mathcal{Y}_r \otimes \mathcal{Y}_r + \mathcal{X}_r \otimes \mathcal{X}_r\right) \\ &+ \frac{1}{4} \sum_{\substack{(r,s) \in \{(23)(56),\\ (24)(57),(34)(67)\}}} \left(\mathcal{Y}_r \otimes \mathcal{Y}_s - \mathcal{X}_r \otimes \mathcal{X}_s + \mathcal{Y}_s \otimes \mathcal{Y}_r - \mathcal{X}_s \otimes \mathcal{X}_r\right) \ , \end{aligned} \tag{D3}$$

with

$$w_3 - w_5 = \frac{1}{\sqrt{2}}\mathrm{diag}\,(0,1,0,-1,-1,0,1,0) \ , \ w_4 + w_6 - 2w_8 \ = \ \frac{1}{\sqrt{2}}\mathrm{diag}\,(0,1,-2,1,-1,2,-1,0) \ . \tag{D4}$$

Similarly, for operators acting on quark-anti-quark or anti-quark-quark sites,

$$\sum_a \widetilde{Q}^{(a)} \otimes \widetilde{\bar{Q}}^{(a)} = \sum_a \widetilde{\bar{Q}}^{(a)} \otimes \widetilde{Q}^{(a)} = -\frac{1}{2}(w_3 - w_5) \otimes (w_3 - w_5) - \frac{1}{6}(w_4 + w_6 - 2w_8) \otimes (w_4 + w_6 - 2w_8)$$
$$+ \frac{1}{4} \sum_{\substack{r \in \{(23),(24),(34),\\ (56),(57),(67)\}}} (\mathcal{Y}_r \otimes \mathcal{Y}_r - \mathcal{X}_r \otimes \mathcal{X}_r)$$
$$+ \frac{1}{4} \sum_{\substack{(r,s) \in \{(23)(56),\\ (24)(57),(34)(67)\}}} (\mathcal{X}_r \otimes \mathcal{X}_s + \mathcal{Y}_r \otimes \mathcal{Y}_s + \mathcal{X}_s \otimes \mathcal{X}_r + \mathcal{Y}_s \otimes \mathcal{Y}_r) \quad . \tag{D5}$$

Rewriting these sums in terms of commuting operators gives,

$$\sum_a \widetilde{Q}^{(a)} \otimes \widetilde{Q}^{(a)} = \frac{1}{4}(D^{(12)} \otimes D^{(12)} + C^{(12)} \otimes C^{(12)}) + \frac{1}{4}(D^{(45)} \otimes D^{(45)} + C^{(45)} \otimes C^{(45)})$$
$$+ \frac{1}{4}(D^{(67)} \otimes D^{(67)} + C^{(67)} \otimes C^{(67)}) + \widetilde{Q}^{(3)} \otimes \widetilde{Q}^{(3)} + \widetilde{Q}^{(8)} \otimes \widetilde{Q}^{(8)} , \tag{D6}$$

$$\sum_a \widetilde{\bar{Q}}^{(a)} \otimes \widetilde{\bar{Q}}^{(a)} = \frac{1}{4}(D^{(12)} \otimes D^{(12)} + C^{(12)} \otimes C^{(12)}) + \frac{1}{4}(D^{(45)} \otimes D^{(45)} + C^{(45)} \otimes C^{(45)})$$
$$+ \frac{1}{4}(D^{(67)} \otimes D^{(67)} + C^{(67)} \otimes C^{(67)}) + \widetilde{\bar{Q}}^{(3)} \otimes \widetilde{\bar{Q}}^{(3)} + \widetilde{\bar{Q}}^{(8)} \otimes \widetilde{\bar{Q}}^{(8)}$$
$$= \sum_a \widetilde{Q}^{(a)} \otimes \widetilde{Q}^{(a)} , \tag{D7}$$

$$\sum_a \widetilde{Q}^{(a)} \otimes \widetilde{\bar{Q}}^{(a)} = \frac{1}{4}(D^{(12)} \otimes D^{(12)} - C^{(12)} \otimes C^{(12)}) + \frac{1}{4}(D^{(45)} \otimes D^{(45)} - C^{(45)} \otimes C^{(45)})$$
$$+ \frac{1}{4}(D^{(67)} \otimes D^{(67)} - C^{(67)} \otimes C^{(67)}) + \widetilde{Q}^{(3)} \otimes \widetilde{\bar{Q}}^{(3)} + \widetilde{Q}^{(8)} \otimes \widetilde{\bar{Q}}^{(8)} , \tag{D8}$$

$$\sum_a \widetilde{\bar{Q}}^{(a)} \otimes \widetilde{Q}^{(a)} = \frac{1}{4}(D^{(12)} \otimes D^{(12)} - C^{(12)} \otimes C^{(12)}) + \frac{1}{4}(D^{(45)} \otimes D^{(45)} - C^{(45)} \otimes C^{(45)})$$
$$+ \frac{1}{4}(D^{(67)} \otimes D^{(67)} - C^{(67)} \otimes C^{(67)}) + \widetilde{\bar{Q}}^{(3)} \otimes \widetilde{Q}^{(3)} + \widetilde{\bar{Q}}^{(8)} \otimes \widetilde{Q}^{(8)}$$
$$= \sum_a \widetilde{Q}^{(a)} \otimes \widetilde{\bar{Q}}^{(a)} , \tag{D9}$$

where the $C^{(ij)}$ and $D^{(ij)}$ are given in Eq. (26), and $\widetilde{\bar{Q}}^{(3)} = -\widetilde{Q}^{(3)}$ and $\widetilde{\bar{Q}}^{(8)} = -\widetilde{Q}^{(8)}$ have been used.

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[1] M. C. Bañuls *et al.*, Simulating Lattice Gauge Theories within Quantum Technologies, Eur. Phys. J. D **74**, 165 (2020), arXiv:1911.00003 [quant-ph].

[2] W. Guan, G. Perdue, A. Pesah, M. Schuld, K. Terashi, S. Vallecorsa, and J.-R. Vlimant, Quantum Machine Learning in High Energy Physics, Mach. Learn. Sci. Tech. **2**, 011003 (2021), arXiv:2005.08582 [quant-ph].

[3] N. Klco, A. Roggero, and M. J. Savage, Standard model physics and the digital quantum revolution: thoughts about the interface, Rept. Prog. Phys. **85**, 064301 (2022), arXiv:2107.04769 [quant-ph].

[4] P. Alsing *et al.*, Accelerating Progress Towards Practical Quantum Advantage: The Quantum Technology Demonstration Project Roadmap (2022), arXiv:2210.14757 [quant-ph].

[5] A. Delgado *et al.*, Quantum Computing for Data Analysis in High-Energy Physics, in *Snowmass 2021* (2022) arXiv:2203.08805 [physics.data-an].

[6] C. W. Bauer *et al.*, Quantum Simulation for High-Energy Physics, PRX Quantum **4**, 027001 (2023), arXiv:2204.03381 [quant-ph].

[7] C. W. Bauer, Z. Davoudi, N. Klco, and M. J. Savage, Quantum simulation of fundamental particles and forces, Nature Rev. Phys. **5**, 420 (2023).

[8] D. Beck *et al.*, Quantum Information Science and Technology for Nuclear Physics. Input into U.S. Long-Range Planning, 2023 (2023) arXiv:2303.00113 [nucl-ex].

[9] A. Di Meglio *et al.*, Quantum Computing for High-Energy Physics: State of the Art and Challenges. Summary of the QC4HEP Working Group (2023), arXiv:2307.03236 [quant-ph].

[10] T. D. Cohen, H. Lamm, S. Lawrence, and Y. Yamauchi (NuQS), Quantum algorithms for transport coefficients in gauge theories, Phys. Rev. D **104**, 094514 (2021), arXiv:2104.02024 [hep-lat].

[11] F. Turro, A. Ciavarella, and X. Yao, Classical and quantum computing of shear viscosity for (2+1)D SU(2) gauge theory, Phys. Rev. D **109**, 114511 (2024), arXiv:2402.04221 [hep-lat].

[12] C. W. Bauer, M. Freytsis, and B. Nachman, Simulating Collider Physics on Quantum Computers Using Effective Field Theories, Phys. Rev. Lett. **127**, 212001 (2021), arXiv:2102.05044 [hep-ph].

[13] Z.-Y. Zhou, G.-X. Su, J. C. Halimeh, R. Ott, H. Sun, P. Hauke, B. Yang, Z.-S. Yuan, J. Berges, and J.-W. Pan, Thermalization dynamics of a gauge theory on a quantum simulator, Science **377**, 311 (2022), arXiv:2107.13563 [cond-mat.quant-gas].

[14] E. A. Martinez *et al.*, Real-time dynamics of lattice gauge theories with a few-qubit quantum computer, Nature **534**, 516 (2016), arXiv:1605.04570 [quant-ph].

[15] N. Klco, E. F. Dumitrescu, A. J. McCaskey, T. D. Morris, R. C. Pooser, M. Sanz, E. Solano, P. Lougovski, and M. J. Savage, Quantum-classical computation of Schwinger model dynamics using quantum computers, Phys. Rev. A **98**, 032331 (2018), arXiv:1803.03326 [quant-ph].

[16] C. Kokail, C. Maier, R. van Bijnen, T. Brydges, M. K. Joshi, P. Jurcevic, C. A. Muschik, P. Silvi, R. Blatt, C. F. Roos, and P. Zoller, Self-verifying variational quantum simulation of lattice models, Nature **569**, 355 (2019), arXiv:1810.03421 [quant-ph].

[17] H.-H. Lu, N. Klco, J. M. Lukens, T. D. Morris, A. Bansal, A. Ekström, G. Hagen, T. Papenbrock, A. M. Weiner, M. J. Savage, and P. Lougovski, Simulations of Subatomic Many-Body Physics on a Quantum Frequency Processor, Phys. Rev. A **100**, 012320 (2019), arXiv:1810.03959 [quant-ph].

[18] N. Klco, J. R. Stryker, and M. J. Savage, SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers, Phys. Rev. D **101**, 074512 (2020), arXiv:1908.06935 [quant-ph].

[19] F. M. Surace, P. P. Mazza, G. Giudici, A. Lerose, A. Gambassi, and M. Dalmonte, Lattice gauge theories and string dynamics in Rydberg atom quantum simulators, Phys. Rev. X **10**, 021041 (2020), arXiv:1902.09551 [cond-mat.quant-gas].

[20] A. Mil, T. V. Zache, A. Hegde, A. Xia, R. P. Bhatt, M. K. Oberthaler, P. Hauke, J. Berges, and F. Jendrzejewski, A scalable realization of local U(1) gauge invariance in cold atomic mixtures, Science **367**, 1128 (2020), arXiv:1909.07641 [cond-mat.quant-gas].

[21] B. Yang, H. Sun, R. Ott, H.-Y. Wang, T. V. Zache, J. C. Halimeh, Z.-S. Yuan, P. Hauke, and J.-W. Pan, Observation of gauge invariance in a 71-site Bose–Hubbard quantum simulator, Nature **587**, 392 (2020), arXiv:2003.08945 [cond-mat.quant-gas].

[22] A. Ciavarella, N. Klco, and M. J. Savage, Trailhead for quantum simulation of SU(3) Yang-Mills lattice gauge theory in the local multiplet basis, Phys. Rev. D **103**, 094501 (2021), arXiv:2101.10227 [quant-ph].

[23] Y. Y. Atas, J. Zhang, R. Lewis, A. Jahanpour, J. F. Haase, and C. A. Muschik, SU(2) hadrons on a quantum computer via a variational approach, Nat. Commun. **12**, 6499 (2021), arXiv:2102.08920 [quant-ph].

[24] S. A Rahman, R. Lewis, E. Mendicelli, and S. Powell, SU(2) lattice gauge theory on a quantum annealer, Phys. Rev. D **104**, 034501 (2021), arXiv:2103.08661 [hep-lat].

[25] G. Mazzola, S. V. Mathis, G. Mazzola, and I. Tavernelli, Gauge-invariant quantum circuits for U(1) and Yang-Mills lattice gauge theories, Phys. Rev. Res. **3**, 043209 (2021), arXiv:2105.05870 [quant-ph].

[26] W. A. de Jong, K. Lee, J. Mulligan, M. Płoskoń, F. Ringer, and X. Yao, Quantum simulation of nonequilibrium dynamics and thermalization in the Schwinger model, Phys. Rev. D **106**, 054508 (2022), arXiv:2106.08394 [quant-ph].

[27] W. Gong, G. Parida, Z. Tu, and R. Venugopalan, Measurement of Bell-type inequalities and quantum entanglement from Λ-hyperon spin correlations at high energy colliders, Phys. Rev. D **106**, L031501 (2022), arXiv:2107.13007 [hep-ph].

[28] H. Riechert, J. C. Halimeh, V. Kasper, L. Bretheau, E. Zohar, P. Hauke, and F. Jendrzejewski, Engineering a U(1) lattice gauge theory in classical electric circuits, Phys. Rev. B **105**, 205141 (2022), arXiv:2108.01086 [cond-mat.mes-hall].

[29] N. H. Nguyen, M. C. Tran, Y. Zhu, A. M. Green, C. H. Alderete, Z. Davoudi, and N. M. Linke, Digital Quantum Simulation of the Schwinger Model and Symmetry Protection with Trapped Ions, PRX Quantum **3**, 020324 (2022), arXiv:2112.14262 [quant-ph].

[30] A. N. Ciavarella and I. A. Chernyshev, Preparation of the SU(3) lattice Yang-Mills vacuum with variational quantum methods, Phys. Rev. D **105**, 074504 (2022), arXiv:2112.09083 [quant-ph].

[31] M. Illa and M. J. Savage, Basic elements for simulations of standard-model physics with quantum annealers: Multigrid and clock states, Phys. Rev. A **106**, 052605 (2022), arXiv:2202.12340 [quant-ph].

[32] J. Mildenberger, W. Mruczkiewicz, J. C. Halimeh, Z. Jiang, and P. Hauke, Probing confinement in a $\mathbb{Z}_2$ lattice gauge theory on a quantum computer (2022), arXiv:2203.08905 [quant-ph].

[33] A. Ciavarella, N. Klco, and M. J. Savage, Some Conceptual Aspects of Operator Design for Quantum Simulations of Non-Abelian Lattice Gauge Theories (2022) arXiv:2203.11988 [quant-ph].

[34] S. A Rahman, R. Lewis, E. Mendicelli, and S. Powell, Self-mitigating Trotter circuits for SU(2) lattice gauge theory on a quantum computer, Phys. Rev. D **106**, 074502 (2022), arXiv:2205.09247 [hep-lat].

[35] M. Asaduzzaman, S. Catterall, G. C. Toga, Y. Meurice, and R. Sakai, Quantum simulation of the N-flavor Gross-Neveu model, Phys. Rev. D **106**, 114515 (2022), arXiv:2208.05906 [hep-lat].